

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	[Enter your Team ID]
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Project Setup	USN-1	As a developer, I can set up the FastAPI project, virtual environment, folder structure and Gemini API key configuration.	3	High	Team Member 1
	User Input	USN-2	As a user, I can enter my name, user ID, age, weight, fitness goal and workout intensity in a form on the home page.	3	High	Team Member 4
		USN-3	As a user, my details are saved in the SQLite database so I do not have to re-enter them.	2	High	Team Member 3
	Workout Plan Generation	USN-4	As a user, I can generate a personalized 7-day workout plan for my goal and intensity using Gemini Pro.	5	High	Team Member 1
		USN-5	As a user, I can see each day's plan as warm-up, main workout and cooldown.	3	High	Team Member 1
		USN-6	As a user, my generated plan is stored in the database.	2	High	Team Member 3
		USN-7	As a user, I can view my plan and details on the result page.	2	High	Team Member 4
Sprint-2	Nutrition Tips	USN-8	As a user, I can get a nutrition or recovery tip based on my fitness goal using Gemini Flash.	3	High	Team Member 1
		USN-9	As a user, I can see the nutrition tip below my workout plan.	2	Medium	Team Member 4
	Feedback & Plan Update	USN-10	As a user, I can submit feedback (for example, "more focus on cardio") using my user ID.	3	High	Team Member 4
		USN-11	As a user, my plan is regenerated using my feedback with Gemini Pro.	5	High	Team Member 1
		USN-12	As a user, my original and updated plans are stored separately so both versions are kept.	3	High	Team Member 3
		USN-13	As a user, I see a confirmation message after my plan is updated.	1	Medium	Team Member 4
	Error Handling	USN-14	As a user, I see a clear message if my user ID is not found or the AI service fails.	3	Medium	Team Member 2
Sprint-3	Admin Dashboard	USN-15	As an admin, I can view all registered users and their details.	3	High	Team Member 2
		USN-16	As an admin, I can compare each user's original and updated plans in a table.	3	High	Team Member 2
		USN-17	As an admin, I can delete a user from the dashboard.	2	Medium	Team Member 3
		USN-18	As an admin, I must log in before I can open the all-users page.	4	Medium	Team Member 2
	User Interface	USN-19	As a user, I can use a gym-themed, responsive home page on desktop and mobile.	3	Medium	Team Member 4

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
		USN-20	As a user, I can read my plan, tip and feedback form on clean, styled Jinja2 pages.	3	Medium	Team Member 4
		USN-21	As a user, I can move between the home, feedback and admin pages using simple links.	2	Low	Team Member 4
Sprint-4	Testing	USN-22	As a developer, I can test every route using FastAPI's /docs page.	3	High	Team Member 2
		USN-23	As a developer, I can test plans for different goals and intensities to check quality.	3	High	Team Member 1
		USN-24	As a developer, I can fix bugs in the feedback and database flow.	3	Medium	Team Member 3
	Local Deployment	USN-25	As a developer, I can run the app locally with Uvicorn at 127.0.0.1:8000.	3	High	Team Member 2
		USN-26	As a developer, I can keep the Gemini API key safely in a .env file.	2	High	Team Member 1
	Documentation & Demo	USN-27	As a team, we prepare the project report with screenshots of each page.	3	Medium	Team Member 4
		USN-28	As a team, we run a final demo and verify all four scenarios.	3	Medium	Team Member 2

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	17 Feb 2025	22 Feb 2025	20	22 Feb 2025
Sprint-2	20	6 Days	24 Feb 2025	01 Mar 2025	20	01 Mar 2025
Sprint-3	20	6 Days	03 Mar 2025	08 Mar 2025	20	08 Mar 2025
Sprint-4	20	6 Days	10 Mar 2025	15 Mar 2025	20	15 Mar 2025

Total: 28 user stories, 80 story points over 4 sprints.

Velocity:

Each sprint is 6 days long and the team's velocity is 20 story points per sprint. The average velocity (AV) per iteration unit (story points per day) is:

AV = velocity / sprint duration = 20 / 6 = 3.33 story points per day

Burndown Chart:

A burndown chart shows the work left to do versus time. For FitBuddy, 80 story points are planned across 24 days (4 sprints of 6 days). The dashed line is the ideal burndown; the blue line is the actual progress (sample data, update with your real daily progress).

